

Nirma University
Institute of Technology
Semester End Examination (IR), January-2023
Master of Computer Applications, Semester - 1
6CS105 Computer Networks

Roll/
Exam No.

Supervisor's Initial
with date

Time: 3 Hours

Max Marks: 100

Instructions:

1. Attempt all questions. (No Optional Questions)
2. Figures to the right indicate full marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

SECTION - I

Q-1. Answer the following questions. [16]

(A) Discuss OSI reference model in detail. Describe the functions of each layer and draw needed diagram. (8)
CO1 BL2

(B) Why Selective-Repeat ARQ (Automatic Repeat Request) protocol gives better performance as compared to Go-Back-N ARQ? Justify your answer with proper diagram and critical comments. (8)
CO2 BL5

Q-2. Answer the following questions. [18]

(A) Distinguish between unicast, multicast and broadcast. (6)
CO1 BL2

(B) A 7-bit binary value 1010111 (data-word) is to be encoded using an even-parity Hamming code. What is the binary value (code-word) after encoding? (6)
CO2 BL3

(C) Discuss key characteristics of Fiber Optic cable with suitable diagram. Also write advantages and disadvantages for the same. (6)
CO1 BL2

Q-3. Answer the following questions. [16]

(A) How many cables and ports are needed, if there are 5 devices in mesh topology? Justify your answer with necessary calculation. (4)
CO1 BL3

(B) Illustrate with example the concept of bit stuffing used in framing. (6)
CO2 BL4

(C) Give the comparative analysis of the functionalities of p-persistent and non-persistent CSMA (Carrier Sense Multiple Access) protocol. (6)
CO2 BL4

SECTION – II

Q-4. Answer the following questions.**[16]**

- (A) Justify the requirement of minimum frame length in IEEE 802.3. (4)
CO2 BL5
- (B) Find the error in the following IPv4 addresses and mention the same with reason. (4)
CO3 BL3
- (i) 112.58.048.79
 - (ii) 212.43.5.4.11
 - (iii) 82.305.12.34
 - (iv) 32.11001100.43.23
- (C) One of the addresses in a block is 188.200.110.81/27. Find the network mask, number of addresses in the block, the first address and the last address of the block. (8)
CO3 BL3

Q-5. Answer the following questions.**[18]**

- (A) Compare and contrast between Bridge and Router. (6)
CO2 BL4
- (B) Which protocol is used for communication between web server and web client in the Internet? Discuss salient features of the protocol. (6)
CO2 BL2
- (C) Explain in detail: User Datagram header format (6)
CO2 BL2

Q-6. Answer the following questions.**[16]**

- (A) Explain in detail the TCP connection establishment phase using flow diagram. (8)
CO2 BL2
- (B) Do you think FTP and Telnet services have to be part of network protocol suit? Justify your answer with suitable scenario. (8)
CO2 BL5